Paper Code : 21325
F-425
B.C.A. (Fifth Semester)
Examination, 2023-24
(New Course)

Paper - BCA-505-N

ADVANCED COMPUTER ARCHITECTURE

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) What are data hazards that are noticed in instruction level parallelism? Explain dynamic scheduling for overcoming these hazards.

(b) What is associative memory? Explain its advantages. 7+7

2. (a) An analysis requires multiplication of very large matrices. Which type of computer will you suggest for them? Give reason in support of you answer.

(b) Differentiate between data flow computer and control flow computer. 7+7

3. What is the major hurdle of pipelining? How do they reduce the performance? What are the techniques to be used to improve the performance? 14

4. What are the five primary approaches, in use, for multiple issue processors? Give the primary characteristics that distinguish them. 14

5. What is multiprocessor cache coherence? Illustrate the problem and show how different processors have different values for the same location. 14

6. (a) What is distributed memory multicomputer?

   (b) Explain SIMD computers? 7+7

7. Explain these terms with diagrams (if any) : 14

   (i) RISC scaler processors

   (ii) Loosely coupled multiprocessors

   (iii) UMA

   (iv) MUMA